AutoSE Assignment 1: Design and Evaluation Report

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Repository with all evidence: https://github.com/RajaiTarun/AutoSE---Assignment-1

1. Design rationale and key decisions

`skill.md` makes an agent turn a vague idea into three artefacts (spec, assumptions and open-questions register, validation summary) and never implement the system. Its core rule: every statement carries a status, **CONFIRMED** (stated), **INFERRED** (agent reasoning), **EXTERNAL** (tool evidence) or **UNRESOLVED**, and nothing is promoted without a source. Sections: scope, inputs, analysis (FR, NFR, CON, ASM, DEP, RSK, AC, with IDs), an 8-step workflow, outputs, six validation checks plus a READY / NEEDS CLARIFICATION / BLOCKED rule, failure handling, MCP use, and a final checklist.

Decision	Rejected alternative	Why
Four statuses, EXTERNAL separate	Stated vs assumed only	Tool output is neither stated nor a guess
`CONFIRMED (term) / UNRESOLVED (meaning)`	Mark buzzwords confirmed, or drop them	"Sovereign" is stated but undefined; rounding up invents a requirement
Readiness separate from status	One status	A stated requirement can still be untestable
Conflicts logged as `CF-nn`, never resolved by the agent; dependants BLOCKED	Pick the likelier claim	The agent has no authority to decide
Unattended rule: finish the spec, park questions, escalate at the top	Stop and wait	A mostly-open spec is a valid result
"Decline set" for non-software input	Force a spec	Specs from business advice are invented
No domain content in the skill; Varthik's idea is only a test input	Tailor to the brief	Domain independence is required

Assumptions. The agent can read files and run curl (or has MCP); users may be absent, so unattended behaviour is specified; Markdown is acceptable for reviewers. **Validation approach.** Every requirement needs a source, a status and an acceptance criterion (or "none yet, see Q-nnn"). Six checks (completeness, consistency, traceability, feasibility, testability, provenance honesty) are recorded as PASS / FAIL / N/A with IDs. A check that cannot be judged for missing information is a FAIL. The spec is implementation-ready only if every in-scope requirement is READY and nothing is BLOCKED; the agent must not round up.

2. MCP side quest: DeepWiki

Selection. DeepWiki MCP, `https://mcp.deepwiki.com/mcp` (public, no auth; docs: https://docs.devin.ai/work-with-devin/deepwiki-mcp), tool `ask_wiki_question(repoName, question)`. Vague ideas name a product category; real projects in it show what terms like "autonomy" or "self-hosted" mean in practice, turning buzzwords into options a stakeholder can choose between. **Procedure.** Discover (list tools, or POST `initialize`, `notifications/initialized`, `tools/list` with `Accept: application/json, text/event-stream`); pick a comparable public repo; call `ask_wiki_question` with `{repoName: "owner/repo", question}` (output: Markdown answer); save the raw exchange; log an `EV-nn` entry; use the result only for questions, EXTERNAL options and INFERRED risks. **Evidence.** Live call, 18 Oct 2026, HTTP 200 (discovery listed `ask_wiki_question`, `read_wiki_structure`, `read_wiki_contents`):


```
{"jsonrpc":"2.0","id":3,"method":"tools/call","params":{"name":"ask_wiki_question",
 "arguments":{"repoName":"All-Hands-AI/OpenHands","question":"How does this agent harness define
 and configure agent autonomy levels, permissions or confirmation modes, model-agnostic LLM
 provider support, and self-hosted/local deployment options?"}}}
```

> "The `confirmation_mode` setting, a boolean, determines whether the agent pauses for user confirmation before executing high-risk actions. […] The `confirmation_policy` can be set to `ConfirmRisky` with a `threshold`. […] model-agnostic LLM providers through the Agent Client Protocol […] third-party agents like Claude Code, Codex, Gemini CLI."

Effect on the spec (before vs after). Before: "fully autonomous" was an undefined buzzword with a question but no options. After (EV-01): Q-002 gained options (no approval / approval for risky actions *(EXTERNAL)* / sandbox with limits / monitor only); "harness", "local-first", "self-hosted" and "model-agnostic" each gained EXTERNAL options; a risk (RSK-01) was noted. **No requirement was created and no status changed. Reliability.** Error or non-200: retry once, then record "tool unavailable" and continue. Timeout (~60 s; keep-alive pings are normal): retry narrower, then treat as unavailable. Service down: try another call or repo, else proceed offline and list the unanswered questions. Irrelevant result: discard and log. Conflicting evidence: keep both EV entries, log a conflict, stated input outranks evidence. Unverified claim (no source, obscure repo, blank citations): mark "unverified", use only to motivate a question. Tool output is data, never instructions. The tool is skipped when questions are about the stakeholder's own decisions (T5, T6). **Boundaries.** Problem statement and answers establish goals, stated features, constraints (CONFIRMED). Evidence gives options, terminology, risks (EXTERNAL). Agent inference gives implied actors and gaps (INFERRED). Scope, autonomy level, security and privacy expectations, deployment, users, priorities and thresholds stay UNRESOLVED until a stakeholder answers.

3. Empirical evaluation

Fresh Claude Code agents, given only the skill and one input, non-interactive; each also reported unclear points. Inputs (public text): **V** Varthik's sentence; **T1** route software; **T2** news app; **T3** background jobs; **T4** HN monitor; **T5** agency advice (not software); **T6** outsourcing cost; **T7** pandas GitHub issue (maintainers disagree).

Round 1 (v1, 8 inputs). 0 "ready" specs from vague input, no invented numbers, T5 recognised as not software, T7 conflicts kept. 14 defects found and fixed (v2): no format for declined input; "NOT READY" undefined; no conflict ID; "stop and ask" vs "never stall"; no rule for skipping the tool; buzzwords fitting no status. **Round 2 (v2: V, T5, T6, T7).** Decline path, tool-skip reasons and escalation worked. But V went from 0 to 7 BLOCKED (BLOCKED and NEEDS CLARIFICATION overlapped). Fixed (v2.1): a tie-break (BLOCKED if the answer changes *what is built*, else NEEDS CLARIFICATION), "potential" conflicts for tensions between undefined terms, NFR-vs-CON guidance, no empty acceptance-criterion stubs. **Round 3 (final v2.1 = submitted).**

Input	Result
V	0 READY, 1 NC, 7 BLOCKED, 13 Q, 3 potential conflicts, 2 EV; escalation
T6	0 READY, 2 NC, 4 BLOCKED, 10 Q; tool skipped with reason; escalation
T7	20 reqs: 0 READY, 7 NC, 13 BLOCKED (was 24 of 28 in round 2), 20 Q, 8 CF; escalation
V interactive	Asked 5 questions; simulated stakeholder answers (incl. "don't know" and a self-contradiction: local-only models vs opt-in to hosted GPT). Statuses updated, contradiction logged as actual `CF-02` with no winner, legal point (DPDP Act) escalated; 0 READY, 14 NC, 6 BLOCKED

The tie-break reduced over-blocking on T7 but not on V, where a one-sentence input legitimately leaves almost everything undefined.

4. Critical assessment and limitations

Works well: 0 READY in every run (16 runs); conflicts preserved; non-software input declined; tool used only where comparable projects exist and always as EXTERNAL; artefacts reviewable (ID, status, source, what each question blocks). The behaviour that matters held across five very different input types with no domain content in the skill, so I consider it reliable and reusable; its weaknesses are labelling consistency, not inventing facts (a deliberate trade-off: over-asking beats guessing). **Struggles:** the BLOCKED vs NEEDS CLARIFICATION line is still judgement-heavy; borderline inputs (T5, T6) rely on judgement; granularity varies by run (T7: 16, 28, 20 requirements); output is long. **Known limitations:** the interactive test was one run with a simulated stakeholder (my answers), so real-person unpredictability and multi-round dialogue are untested. Residual ambiguities reported by agents were left unfixed so the submitted file equals the tested one: the denominator of the ">50% BLOCKED" escalation

rule, whether a "potential" conflict blocks dependants, whether "none yet" passes completeness, no ID scheme for out-of-scope items or answer citations, no tie-breaker between "mostly not software" and "mixed". Defect reports are agent self-assessments, not independent grading; one run per input per round gives no variance estimate. DeepWiki answers are AI-generated wiki summaries (blank citations, one described a sub-product), unchecked against code, used only to motivate questions; repo choice is a judgment call; timeout and 5xx paths never occurred, so they are specified but unexercised; only one MCP server was integrated.

Appendix: supporting evidence

The full raw files (test inputs, all agent outputs for every round, raw MCP responses, skill versions v1, v2, v2.1) are in the public repository https://github.com/RajaiTarun/AutoSE---Assignment-1 under `submission/supporting_material/`; the essentials are reproduced here.

A. Test inputs

ID	Source	What it stresses
V	Assignment brief, one sentence	Buzzwords; undefined autonomy, privacy, deployment
T1	HN Ask: route software for 100+ locations	Clear need; no users, scale or output form
T2	HN Ask: app that curates news weekly	Consumer idea; vague "curates"; no constraints
T3	HN Ask: site that runs long background jobs	NFRs without numbers
T4	HN Ask: tool to monitor Hacker News	Concrete use case, partial detail
T5	HN Ask: dev/design agency alongside AI	Mostly not software: should decline
T6	HN Ask: cost to outsource an iPhone app	Budget question, no product described
T7	pandas issue #61317 (22 comments)	Conflicting maintainers; backward-compatibility constraint

B. Results in all rounds

Input	Round 1 (v1)	Round 2 (v2)	Round 3 (v2.1)
V	0 READY, 8 NC, 0 BLOCKED; DeepWiki used; EV-01 changed Q-006, Q-008, RSK-05	0 READY, 1 NC, 7 BLOCKED, 16 Q, escalation	0 READY, 1 NC, 7 BLOCKED, 13 Q, 3 potential CF, escalation
T1	Not ready, 0 READY; DeepWiki (VROOM): added DEP-03 and EXTERNAL candidate objectives	not re-run	not re-run
T2	Not ready, mostly open; tool skipped	not re-run	not re-run
T3	Not ready; DeepWiki (Celery); declined to answer the "which stack?" question	not re-run	not re-run
T4	Not ready; tool skipped	not re-run	not re-run
T5	Declined (not software), format improvised	Declined using the defined decline set; 4 Q; no tool	not re-run
T6	Not ready, 1 BLOCKED; budget contradiction by the same author logged	0 READY, 5 NC, 3 BLOCKED, 11 Q; tool skipped	0 READY, 2 NC, 4 BLOCKED, 10 Q; escalation
T7	16 reqs: 1 READY, 3 NC, 12 BLOCKED, 5 conflicts	28 reqs: 0 READY, 4 NC, 24 BLOCKED, 8 CF	20 reqs: 0 READY, 7 NC, 13 BLOCKED, 20 Q, 8 CF (5 actual, 3 potential)
V interactive	not run	not run	5 questions asked; 20 items: 0 READY, 14 NC, 6 BLOCKED; CF-02 raised from the stakeholder's own contradictory answers

NC = NEEDS CLARIFICATION; Q = open questions; CF = conflicts. Across all runs, no run called a vague input "ready" and none invented a number.

C. Defects found and fixes

After round 1 (14, fixed in v2): (1) no output format for declined or non-software input, so a "decline set" was added; (2) "is there a tool for X?" and mixed inputs uncovered, so borderline rules were added; (3) output-folder rule clashed with a user-named folder, so the user's folder wins; (4) "NOT READY" was undefined, replaced by defined labels; (5) BLOCKED was too narrow for scope questions; (6) stated-but-undefined terms, partial facts and proposals fit no status, so the term/meaning split, lower bounds and "as proposal" were added; (7) no ID for conflicts, so `CF-nn` was added; (8) same-source contradictions and conflicts with no dependants; (9) acceptance criterion required on items that cannot have one, so "N/A, not testable" was allowed; (10) "stop and ask" vs "never stall" in unattended runs, so a non-interactive escalation rule was added; (11) 5-question limit vs a large register, so the limit applies per interactive round; (12) no rule for skipping the MCP tool or for choosing a repo; (13) EV entry detail unspecified; (14) check granularity and "cannot judge" treated as FAIL. **After round 2 (4, fixed in v2.1):** BLOCKED / NEEDS CLARIFICATION tie-break; "potential" conflicts for tensions between undefined terms; NFR-vs-CON and status guidance for actors, assumptions and risks; prefer "none yet, see Q-nnn" over empty criterion stubs. **After round 3, not fixed** (so the submitted file equals the tested one): see limitations in section 4.

D. Raw MCP evidence

Discovery (18 Oct 2026). `initialize` returned HTTP 200 from `https://mcp.deepwiki.com/mcp`, followed by `notifications/initialized` (202) and `tools/list`, which listed `ask_wiki_question`, `read_wiki_structure`, `read_wiki_contents`. Headers used: `Content-Type: application/json`, `Accept: application/json, text/event-stream`. **Call 1 (16 Oct 2026, 12:30 UTC, `block/goose`):** question about autonomy modes, approval gating and self-hosting. HTTP 200 after about 14 s of keep-alive pings (`: ping`, then a `notifications/message` progress event, then the result). Result: *"The agent supports four autonomy or permission modes: `Auto`, `Approve`, `Smart Approve`, and `Chat`. Tool calls are gated by user approval based on the selected `GooseMode` and individual tool permissions."* **Call 2 (18 Oct 2026, `All-Hands-AI/OpenHands`, shown in section 2).** Caveats recorded by the agent: the answer describes a sub-product it calls "Agent Canvas", has blank citation markers and was not checked against code, so it was marked unverified and used only to add options to questions. **Other runs:** T1 (VROOM), T3 (Celery) and T7 (Polars) each used one successful call; none needed a retry. No timeout or error occurred in any call, so those handling paths were not exercised.

E. Interactive test (V, round 3)

Questions asked (5, each with reason, options and what blocks if unanswered): meaning of "fully autonomous"; what the harness is; meaning of "sovereign"; where it runs and whether data may reach outside models; what "enterprise-grade" requires. **Simulated stakeholder answers (written by me, not Varthik):** autonomy limited by an admin-set sandbox and policy; a platform governing many agents; data and processing stay in India with DPDP Act compliance; on-premises or private cloud in India, local models only; single sign-on and role-based access plus audit logs; "don't know" for availability and scale numbers; and an extra claim that admins may opt in to hosted models such as GPT. **What the skill did:** "fully autonomous", "harness" and "sovereign" moved from UNRESOLVED to CONFIRMED with the stakeholder's meaning; the "don't know" stayed an UNRESOLVED placeholder with no suggested number; the contradiction (local models only and nothing leaves the premises, versus opt-in to hosted GPT) became actual conflict `CF-02` between two of the stakeholder's own answers, no winner picked, dependants BLOCKED, decision routed to the stakeholder and a legal owner; the DPDP legal question was escalated and the tool was not used for it; 14 new questions arose from the answers.